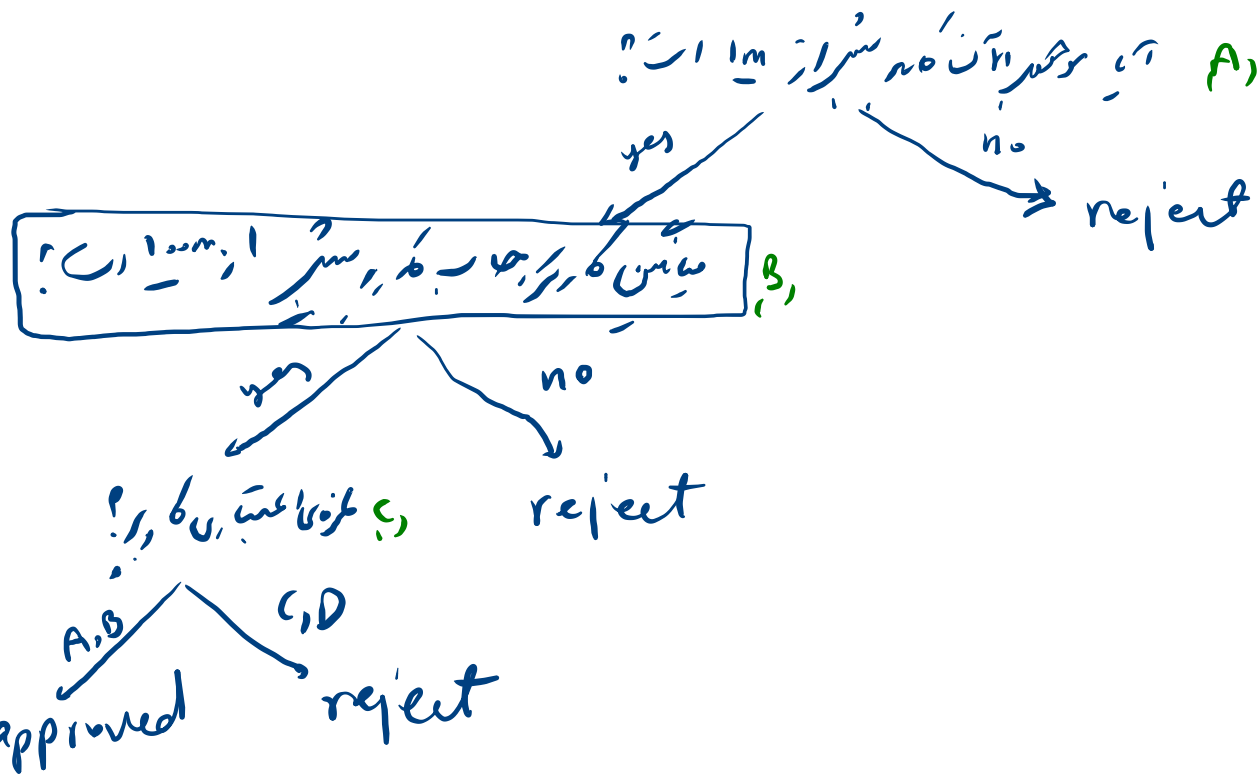


$A \rightarrow B \rightarrow C$

$B \rightarrow A \rightarrow C$

$\vdots$

37595
سند کفیل وام! سند آن به کار برآید ۱۵ وام ملحق ناکرد؟



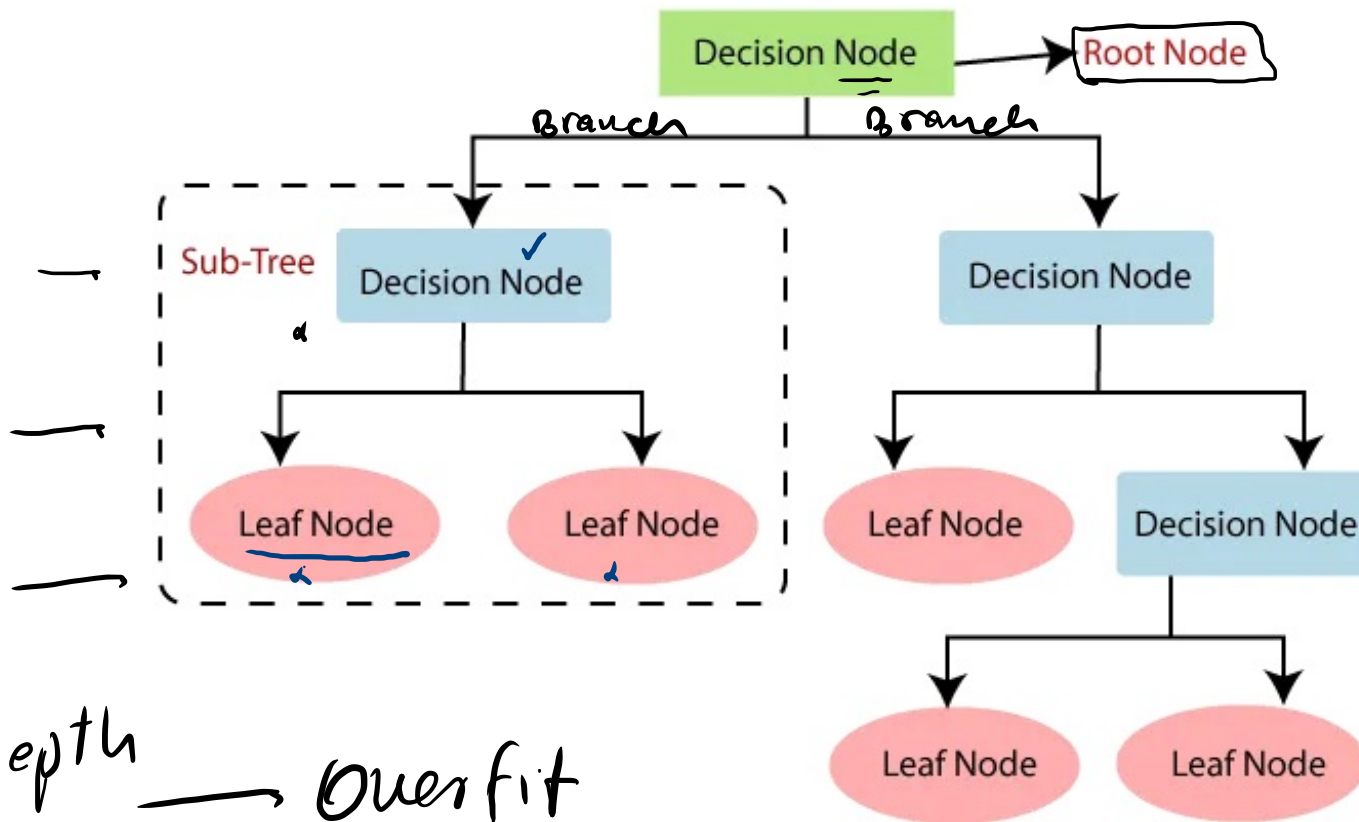
DT

a Decision Tree

دخت لقمه

user → approved ✓
 → reject ✓

depth



high depth → overfit
high branches

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

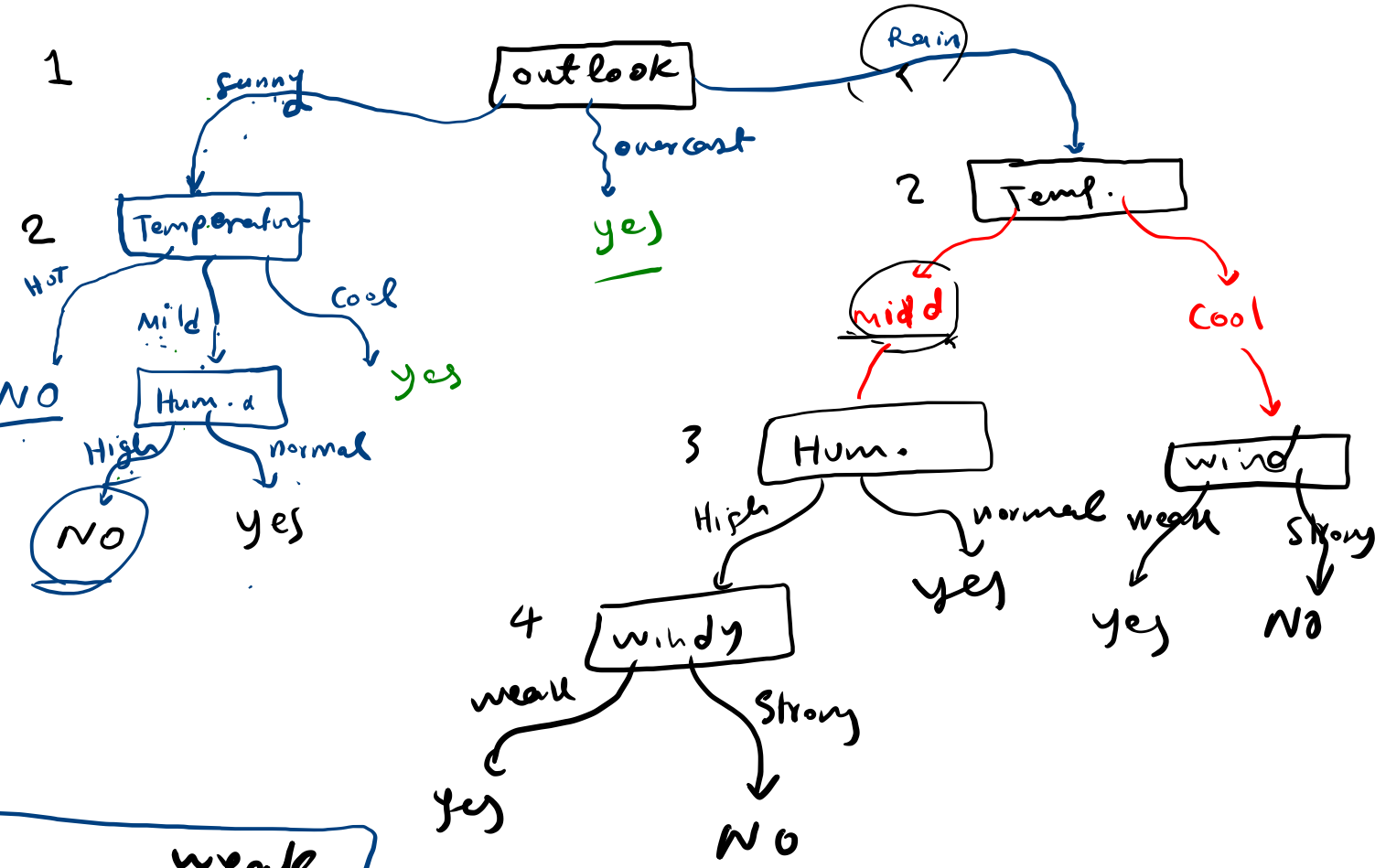
Sunny
overcast
rain

Hot
mild
Cool

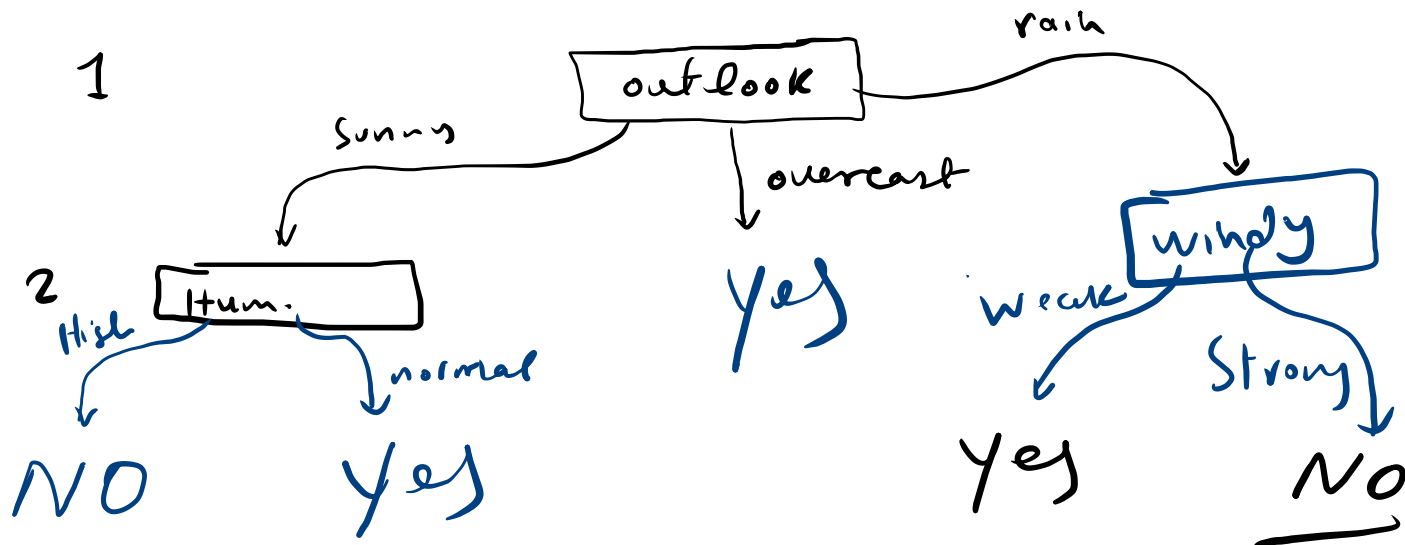
High
normal

Weak
Strong

Sunny mild High weak



Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



سوال : OT گھونہ باہر ترتیب mode ہالان

انتھریٹم کے باکٹریز عمق : جواب دہتی زیریں؟

α

Entropy

$$= - \sum_{i=1}^c p_i \cdot \log_2 p_i$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Entropy : - \sum_{i=1}^c p_i \log_2 p_i$$

$$c : \begin{cases} \text{yes} : + \longrightarrow 9 \longrightarrow P_+ = \frac{9}{14} \\ \text{no} : - \longrightarrow 5 \longrightarrow P_- = \frac{5}{14} \end{cases}$$

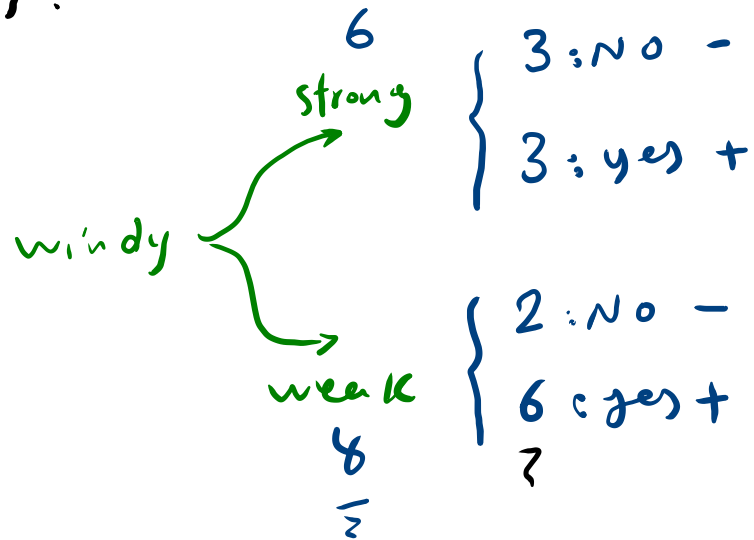
$$Entropy = - [P_+ \log_2 P_+] - [P_- \log_2 P_-] =$$

$$Entropy = - \left[\frac{9}{14} \log_2 \left(\frac{9}{14} \right) \right] - \left[\frac{5}{14} \log_2 \left(\frac{5}{14} \right) \right] = \checkmark \quad \underline{\text{EPT}}$$

Target

$$\text{Gain}(\text{playTennis}, \text{windy}) = \underbrace{\text{Entropy}(\text{playTennis})}_{\text{EPT}} - \underbrace{\sum \frac{|S_v|}{|S|} \text{Entropy}(S_v)}_{\text{SPT}}$$

windy



$$\frac{S_{\text{weak}}}{S} \text{Entropy}(S_{\text{weak}}) +$$

$$\frac{S_{\text{strong}}}{S} \text{Entropy}(S_{\text{strong}})$$

$$\frac{8}{14} \text{Entropy}(S_{\text{weak}}) + \frac{6}{14} \text{Entropy}(S_{\text{strong}})$$

WPT

$$\left[-\frac{2}{8} \log_2\left(\frac{2}{8}\right) - \frac{6}{8} \log_2\left(\frac{6}{8}\right) \right]$$

SPT

$$\left[-\frac{3}{6} \log_2\left(\frac{3}{6}\right) - \frac{3}{6} \log_2\left(\frac{3}{6}\right) \right]$$

$$\text{Gain}(\text{play}_{\text{Tennis}}, \underline{\text{windy}}) =$$

$$E_{PT} - \left[\frac{8}{14} U_{PT} + \frac{6}{14} S_{PT} \right]$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
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D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\text{Gain}(\text{playTennis}, \text{windy}) = A$$

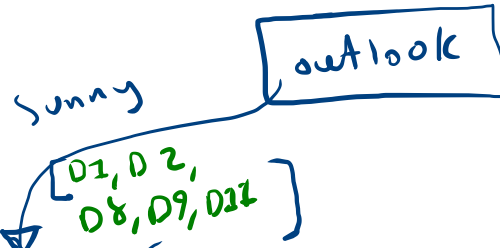
$$\text{Gain}(\text{play Tennis}, \text{Humidity}) = B$$

$$\text{Gain}(\text{play Tennis}, \text{Temp.}) = C$$

$$\text{Gain}(\text{play Tennis}, \text{outlook}) = D$$

$$\max(A, B, C, D) = \text{root node.}$$

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



Temp?
Hum?
wind?

Gain (PT, Temp) : AA

Gain (PT, Hum) : BB

Gain (PT, windy) : CC

max (AA, BB, CC)

max (AA, BB, CC)

D1
D2
D8
D9
D11

Thelma